

HR Analytics Dashboard & Employee Attrition Analysis

Project Objective:

The objective of this project is to analyze employee attrition patterns and identify the key factors influencing workforce turnover within the organization. Using SQL and Power BI, the project aims to transform raw HR data into meaningful business insights that help understand employee behavior, department-wise attrition trends, salary distribution, overtime impact, and overall workforce performance.

This project focuses on building interactive dashboards and analytical reports to assist HR teams and management in making data-driven decisions for improving employee retention, reducing attrition rates, and enhancing organizational productivity.

Key objectives of the project include:

1. Analyzing overall employee attrition rate
2. Identifying departments with high employee turnover
3. Evaluating the impact of overtime on attrition
4. Studying salary and age-group patterns among employees
5. Creating KPI-driven dashboards for HR decision-making
6. Generating actionable insights and business recommendations using data analytics techniques

2. Tools Used

1. SQL
2. MySQL
3. Power BI
4. Excel

Dataset Information

The dataset used in this project is the IBM HR Analytics Employee Attrition dataset. It contains employee-related information that helps analyze workforce trends, employee satisfaction, and attrition behavior within an organization.

The dataset includes various employee attributes such as demographic details, job-related information, compensation, work environment, and attrition status. It is widely used for HR analytics and employee retention analysis projects.

Dataset Source

- IBM HR Analytics Employee Attrition Dataset
- Source: Kaggle

Total Records

- 1470 employee records

Key Features / Columns

The dataset contains multiple columns, including:

1. Age – Employee age
2. Attrition – Indicates whether the employee left the company
3. Department – Employee department
4. JobRole – Employee job designation
5. Gender – Gender of employee
6. MonthlyIncome – Monthly salary of employee
7. OverTime – Whether employee works overtime
8. JobSatisfaction – Satisfaction level of employee
9. YearsAtCompany – Total years worked in company
10. Education – Employee education level
11. BusinessTravel – Travel frequency for work
12. MaritalStatus – Employee marital status

Dataset Purpose

The dataset is used to identify patterns and factors affecting employee attrition. It helps analyze workforce behavior and supports HR teams in making strategic decisions to improve employee retention and organizational performance.

Business Problems Solved:

1. Analyzed overall employee attrition trends
2. Identified departments with highest attrition
3. Evaluated impact of overtime on employee retention
4. Analyzed salary distribution and attrition relationship
5. Identified high-risk employee age groups
6. Studied workforce distribution across job roles

7. Generated KPI-based HR insights for decision making
8. Provided data-driven recommendations to improve employee retention

SQL Analysis:

Query1:Total_emp

```
SELECT COUNT(*) AS total_employees  
FROM employee_attrition;
```

	total_employees
1	1470

Query 2 : Attrition Distribution

```
SELECT Attrition,  
       COUNT(*) AS employee_count  
FROM employee_attrition  
GROUP BY Attrition;
```

	Attrition	employee_count
1	No	1233
2	Yes	237

Query 3 :Attrition Rate KPI

```
SELECT ROUND(  
    SUM(  
        CASE  
            WHEN Attrition = 'Yes' THEN 1  
            ELSE 0  
        END  
    ) * 100.0 / COUNT(*),
```

2

```
) AS attrition_rate  
FROM employee_attrition;
```

	attrition_rate
1	16.12

Insights: Attrition rate KPI was calculated to evaluate overall employee retention performance and measure workforce stability within the organization.

Query 4 : Department-wise Attrition

```
SELECT Department,  
       COUNT(*) AS attrition_count  
FROM employee_attrition  
WHERE Attrition = 'Yes'  
GROUP BY Department  
ORDER BY attrition_count DESC;
```

	Department	attrition_count
1	Research & Development	133
2	Sales	92
3	Human Resources	12

Insights: Department-wise attrition analysis was performed to identify business units experiencing high employee turnover and retention challenges.

Query 5 :Overtime Impact Analysis:

```
SELECT OverTime,  
       COUNT(*) AS employees_left  
FROM employee_attrition  
WHERE Attrition = 'Yes'
```

GROUP BY OverTime;

Insights:

Overtime analysis was conducted to evaluate the relationship between employee workload and attrition behavior.

Query 6 : Average Salary by Department

```
SELECT Department,  
       ROUND(AVG(MonthlyIncome),2) AS avg_salary  
FROM employee_attrition  
GROUP BY Department  
ORDER BY avg_salary DESC;
```

	Department	avg_salary
1	Sales	6959.17
2	Human Resources	6654.51
3	Research & Development	6281.25

Insights: Average salary analysis was performed to compare employee compensation across departments and identify salary distribution patterns within the organization.

Query 7 : Salary Category Analysis Using CASE WHEN

```
SELECT EmployeeNumber,  
       MonthlyIncome,  
       CASE  
         WHEN MonthlyIncome >= 15000 THEN 'High Salary'  
         WHEN MonthlyIncome >= 8000 THEN 'Medium Salary'  
         ELSE 'Low Salary'  
       END AS salary_category  
FROM employee_attrition;
```

	EmployeeNumber	MonthlyIncome	salary_category
1	1	5993	Low Salary
2	2	5130	Low Salary
3	4	2090	Low Salary
4	5	2909	Low Salary
5	7	3468	Low Salary
6	8	3068	Low Salary
	10	8450	Medium Salary

Insight: Employees were categorized into salary groups to analyze workforce compensation patterns and identify salary-based employee segments.

Query 8 : Attrition by Salary Category

SELECT

CASE

WHEN MonthlyIncome >= 15000 THEN 'High Salary'

WHEN MonthlyIncome >= 8000 THEN 'Medium Salary'

ELSE 'Low Salary'

END AS salary_category,

COUNT(*) AS attrition_count

FROM employee_attrition

WHERE Attrition = 'Yes'

GROUP BY salary_category

ORDER BY attrition_count DESC;

	salary_category	attrition_count
1	Low Salary	197
2	Medium Salary	35
3	High Salary	5

Insights: Salary category attrition analysis was conducted to evaluate the relationship between employee compensation and workforce turnover.

AGE GROUP ANALYSIS:

Query 9 :Employee Age Group Classification

SELECT Age,

CASE

WHEN Age < 30 THEN 'Young Employees'

WHEN Age BETWEEN 30 AND 45 THEN 'Mid Age Employees'

ELSE 'Senior Employees'

END AS age_group

FROM employee_attrition;

	Age	age_group
25	34	Mid Age Employees
26	53	Senior Employees
27	32	Mid Age Employees
28	42	Mid Age Employees
29	44	Mid Age Employees
30	46	Senior Employees

Insights: Employee age groups were analyzed to understand workforce demographics and identify major employee segments within the organization.

Query 10 : Attrition by Age Group

SELECT

CASE

WHEN Age < 30 THEN 'Young Employees'

WHEN Age BETWEEN 30 AND 45 THEN 'Mid Age Employees'

ELSE 'Senior Employees'

END AS age_group,

COUNT(*) AS attrition_count

FROM employee_attrition

WHERE Attrition = 'Yes'

GROUP BY age_group

ORDER BY attrition_count DESC;

	age_group	attrition_count
1	Mid Age Employees	112
2	Young Employees	91
3	Senior Employees	34

Insights:Age-group attrition analysis was performed to identify employee segments with higher turnover risk and workforce instability.

JOB ROLE ANALYSIS

SELECT JobRole,

COUNT(*) AS attrition_count

FROM employee_attrition

WHERE Attrition = 'Yes'

GROUP BY JobRole

ORDER BY attrition_count DESC;

	JobRole	attrition_count
1	Laboratory Technician	62
2	Sales Executive	57
3	Research Scientist	47
4	Sales Representative	33
5	Human Resources	12
5	Manufacturing Director	10

Insights: Job role attrition analysis was performed to identify employee positions experiencing high turnover and workforce retention challenges.

JOB SATISFACTION ANALYSIS

Query 12 : Attrition by Job Satisfaction

```
SELECT JobSatisfaction,  
       COUNT(*) AS attrition_count  
FROM employee_attrition  
WHERE Attrition = 'Yes'  
GROUP BY JobSatisfaction  
ORDER BY JobSatisfaction;
```

	JobSatisfaction	attrition_count
1	1	66
2	2	46
3	3	73
4	4	52

Insights: Job satisfaction analysis was conducted to understand how employee satisfaction levels influence workforce attrition behavior.

ADVANCED SQL

Query 13: Ranking Departments by Attrition

```
SELECT Department,  
       COUNT(*) AS attrition_count,  
       RANK() OVER(  
         ORDER BY COUNT(*) DESC  
       ) AS dept_rank  
FROM employee_attrition  
WHERE Attrition = 'Yes'
```

GROUP BY Department;

	Department	attrition_count	dept_rank
1	Research & Development	133	1
2	Sales	92	2
3	Human Resources	12	3

Insights: Window functions were used to rank departments based on employee attrition levels and identify high-risk business units.

Query 14 : CTE Analysis

WITH dept_attrition AS (

SELECT Department,

COUNT(*) AS attrition_count

FROM employee_attrition

WHERE Attrition = 'Yes'

GROUP BY Department

)

SELECT *

FROM dept_attrition

WHERE attrition_count > (

SELECT AVG(attrition_count)

FROM dept_attrition

);

	Department	attrition_count
1	Research & Development	133
2	Sales	92

Insights: Common Table Expressions (CTEs) and subqueries were used to identify departments with attrition levels above the organizational average.

Query 15 :Average Monthly Income of Employees Leaving

SELECT ROUND(AVG(MonthlyIncome),2) AS avg_income_attrition

FROM employee_attrition

WHERE Attrition = 'Yes';

	avg_income_attrition
1	4787.09

Insights: Average income analysis of employees leaving the organization was conducted to study compensation-related attrition patterns.

Query 16 : Workforce Distribution by Department

SELECT Department,

COUNT(*) AS total_employees

FROM employee_attrition

GROUP BY Department

ORDER BY total_employees DESC;

	Department	total_employees
1	Research & Development	961
2	Sales	446
3	Human Resources	63

Insights: Workforce distribution analysis was conducted to understand employee allocation across organizational departments.

Query 17 : Average Years at Company

SELECT ROUND(AVG(YearsAtCompany),2) AS avg_years

FROM employee_attrition;

	avg_years
1	7.01

Insights: Employee experience analysis was performed to evaluate workforce stability and average organizational tenure.

Query 18 : Employees Leaving Early

SELECT

CASE

```
    WHEN YearsAtCompany < 3 THEN 'Early Employees'
    ELSE 'Experienced Employees'
END AS employee_type,
```

```
    COUNT(*) AS attrition_count
```

```
FROM employee_attrition
```

```
WHERE Attrition = 'Yes'
```

```
GROUP BY employee_type;
```

	employee_type	attrition_count
1	Early Employees	102
2	Experienced Employees	135

Insights:Employee tenure analysis was conducted to identify attrition patterns among newly joined and experienced employees.

Query 19 :Top Paying Departments

```
SELECT Department,
```

```
    ROUND(AVG(MonthlyIncome),2) AS avg_salary,
```

```
    RANK() OVER(
```

```
        ORDER BY AVG(MonthlyIncome) DESC
```

```
    ) AS salary_rank
```

```
FROM employee_attrition
```

GROUP BY Department;

	Department	avg_salary	salary_rank
1	Sales	6959.17	1
2	Human Resources	6654.51	2
3	Research & Development	6281.25	3

Insights: Window functions were used to rank departments based on average employee compensation levels.

Dashboard Analysis

KPI Cards Analysis

Total Employees

The dashboard identified the total workforce size of the organization and provided a clear overview of employee distribution across departments.

Average Salary

The average salary KPI evaluated overall employee compensation patterns and helped compare salary distribution within departments.

Attrition Rate

The attrition rate analysis measured workforce turnover percentage and highlighted employee retention performance within the organization.

Attrition Count

The dashboard calculated the total number of employees who left the organization and identified overall attrition trends.

Department-wise Attrition Analysis

The department-wise attrition analysis identified departments experiencing higher employee turnover. Research & Development and Sales departments demonstrated comparatively higher attrition trends, indicating the need for improved employee retention strategies.

Department-wise Salary Distribution Analysis

The salary distribution analysis compared average employee salaries across departments and highlighted compensation differences influencing workforce retention and employee satisfaction.

Overtime Impact Analysis

The overtime analysis revealed that employees working overtime showed higher attrition trends, indicating workload pressure and work-life balance as major factors affecting employee retention.

Attrition Analysis by Age Group

The age-group analysis identified younger employees as having comparatively higher attrition rates, suggesting increased career mobility and lower long-term retention among early-career employees.

Job Satisfaction Analysis

The job satisfaction analysis demonstrated that employees with lower satisfaction levels experienced higher attrition trends, highlighting employee engagement and workplace satisfaction as critical retention factors.

Key Business Insights

1. Research & Development department showed the highest employee attrition.
 2. Overtime employees demonstrated increased turnover trends.
 3. Younger employees showed higher attrition risk compared to experienced employees.
 4. Lower job satisfaction strongly influenced employee exits.
 5. Salary differences impacted employee retention patterns within departments.
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Recommendations

1. Improve employee engagement and retention strategies within high-attrition departments.
2. Reduce excessive overtime workload to improve work-life balance.
3. Strengthen career growth opportunities for younger employees.
4. Improve workplace satisfaction and employee experience initiatives.

5. Review compensation strategies to improve workforce stability and retention.
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Conclusion

The HR Analytics Dashboard successfully identified key workforce attrition patterns using SQL, DAX, and Power BI. The project provided valuable business insights related to employee retention, job satisfaction, overtime impact, salary distribution, and departmental attrition trends. The dashboard supports data-driven HR decision-making and workforce management strategies.

DAX Measures Used in Dashboard:

Total Employees

Total Employees =

COUNT(employee_attrition[EmployeeNumber])

Attrition Count

Attrition Count =

CALCULATE(
COUNT(employee_attrition[Attrition]),
employee_attrition[Attrition] = "Yes"
)

Attrition Rate

Attrition Rate =

DIVIDE(
[Attrition Count],
[Total Employees],
0
)

Average Salary

Average Salary =

AVERAGE(employee_attrition[MonthlyIncome])

Employees Left

Employees Left =

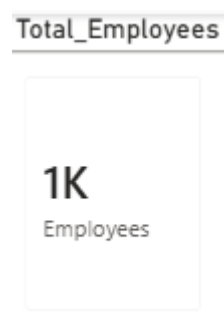
CALCULATE(
COUNT(employee_attrition[EmployeeNumber]),
employee_attrition[Attrition] = "Yes"
)

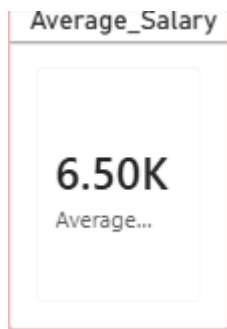
Age Group Column

Age Group =

SWITCH(
TRUE(),
employee_attrition[Age] < 30, "Young Employees",
employee_attrition[Age] <= 45, "Mid Age Employees",
"Senior Employees"
)

Visual Screenshots :





JobRole	Department
JobRole	Department
Healthcare Representative	Human Resources
Human Resources	Research & Development
Laboratory Technician	Sales
Manager	
Manufacturing Director	
Research Director	
Research Scientist	
Sales Executive	
Sales Representative	

Key Insights

1. **Research & Development department showed the highest employee attrition compared to other departments.**
2. **Employees working overtime demonstrated significantly higher turnover trends.**
3. **Younger employees experienced higher attrition rates than experienced employees.**
4. **Lower job satisfaction strongly influenced employee exits and workforce instability.**

5. **Salary distribution differences impacted employee retention patterns across departments.**
 6. **Workforce analytics helped identify critical retention challenges and employee behavior trends within the organization.**
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Overall Project Conclusion

The HR Analytics Dashboard and Employee Attrition Analysis project successfully analyzed workforce attrition patterns using SQL, DAX, and Power BI. The project identified major business factors influencing employee turnover, including overtime, job satisfaction, salary distribution, departmental performance, and employee age groups.

Advanced SQL concepts such as CASE WHEN, CTEs, aggregate functions, GROUP BY, HAVING, and window functions were used to perform detailed workforce analysis. Power BI dashboards and DAX measures were utilized to build interactive visualizations and KPI tracking systems for business reporting.

The project provided meaningful business insights that support data-driven HR decision-making, employee retention strategies, workforce planning, and organizational performance improvement. Overall, the project demonstrated strong analytical, visualization, and business intelligence skills relevant to real-world data analytics roles.